$n = 3$

$$\begin{array}{ccccc} & & 1 & & \\ & 2 & 3 & 2 & \\ 3 & 4 & 5 & 4 & 3 \\ & 2 & 3 & 2 & \\ & & 1 & & \end{array}$$

n = 3,

2 * n + 1

2 * n - 1

```
    *
  *  *  *
*  *  *  *  *
  *  *  *
    *
```

star += 2;
space--;

star -= 2;
space++;

(n=3, star=1, space=2, row=1;)

```
while(row <= 2 * n - 1) {

// space

int i = 1;
while(i <= space) {
  sout(" ");
  i++;
}

star = star - 2;

// star

int j = 1;
while(j <= star) {
  sout("*");
  j++;
}
if(row < n) {
  star += 2;
  space--;
} else {
  star -= 2;
  space++;
}
```

val = 1;

while(row <= 2*n-1){

~~val = 1;~~ ×

row++;

1

2 2 2

3 3 3 3

3 3

2 2 2

1

→ (val ↑) → val++

↓ easy → val--

1

1 1 1

1 1 1 1 1

1 1 1

1

```
- - 1
- 2 3 2
- 2 3 2
3 - 4 - 5 - 4 - 3
2 3 2
1

3 4 5 4 3 // star

j<3

n=3, space=n-1, row=1, star=1, val=1;

while (row <= 2 * n-1) {
    int i=1;
    while(i<=space) {
        sout(" ");
        i++;
    }
    int j=1;
    int number = val;
    while (j<=star) {
        sout(number);
        if(j < star/2 + 1) {
            number++;
        } else {
            number--;
        }
        j++;
    }
    if(row<n) {
        star = star+2;
        space = space - 1;
        val++;
    } else {
        star = star-2;
        space = space+1;
        val--;
    }
    soutln();
    row++;
}
```

Maths Concept →

(extraction of digit)

$\boxed{N = 7789}$

↓

(7, 7, 8, 9)

$7789 \xrightarrow{\% 10} 9,$

$7789 \xrightarrow{/ 10} 778$

$778 \xrightarrow{\% 10} 8$

$778 \xrightarrow{/ 10} 77$

$77 \xrightarrow{\% 10} 7$

$77 \xrightarrow{/ 10} 7$

$7 \xrightarrow{\% 10} 7$

$7 \xrightarrow{/ 10} 0$

10) 7 (0
    0
    ―
    7
    ―

Qus. 1. Sum of digit ⇒

(5487)

```
int sum=0;
while (n>0) {

    int digit = n%10;

    n=n/10;
    sum = sum+digit;
}
```

Dry Run

Sum	n	digit
0	5487	7
7	548	8
15	54	4
19	5	5
24	0	

Sout(sum);

H.W.

n = 587623189

→ Revise the today's class

File Edit Selection View Go Run Terminal Help

Pattern1.java | Pattern2.java | Star_Rhombus.java | MirrorPattern.java | pattern10.java | Pattern_Rombus.java

Pattern_Rombus.java > Pattern_Rombus > main(String[])

```java
public class Pattern_Rombus {
    public static void main(String[] args) {
                    if(j < star/2 + 1){
                        number++;
                    }else number--;        //vertical mirroing done
                    j++;
                    // number++;
                }
                //mirror
                if(row < n){
                    star += 2;
                    space--;
                    val++;     //extra addition
                }else{
                    star -= 2;
                    space++;
                    val--;     //extra addition
                }
                System.out.println();
                row++;
            }
            sc.close();
        }
}
```

Pattern_Rombus.java > Pattern_Rombus > main(String[])

```java
import java.util.Scanner;

public class Pattern_Rombus {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int row = 1,space = n-1,star = 1,val = 1;

        while(row <= 2*n-1){
            //space
            int i = 1;
            while(i <= space){
                System.out.print(s: "\t");
                i++;
            }

            //star
            int j = 1;
            // row start point is hidden on the val , save it on a number
            int number = val;
            while(j <= star){
                // System.out.print(val + "\t");
                System.out.print(number + "\t");

                if(j < star/2 + 1){
                    number++;
```

Lecture_10

ıbus.java | MirrorPattern.java | pattern10.java | Pattern_Rombus.java

Pattern_Rombus.java > Pattern_Rombus > main(String[])

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\91950\Desktop\coding_blocks\Lecture_10> cd "c:\Users\91950\Desktop\coding_blocks\Lecture_10\" ; if ($?) { javac Pattern_Rombus.java } ; if ($?) { java Pattern_Rombus }
5
                                1
                        2       3       2
                3       4       5       4       3
        4       5       6       7       6       5       4
5       6       7       8       9       8       7       6       5
        4       5       6       7       6       5       4
                3       4       5       4       3
                        2       3       2
                                1
PS C:\Users\91950\Desktop\coding_blocks\Lecture_10>
```

Run Testcases ⊗ 0 ⚠ 0 Java: Ready Spaces: 4 UTF-8 CRLF { } Java Go Live Prettier

Fibonacci Series

$n = 11$

0 1 1 2 3 5 8 13 21 34 55

$$\boxed{Fib(n) = fib(n-1) + fib(n-2);}$$

$fib(11) = fib(10) + fib(9)$

$fib(5) = fib(4) + fib(3)$

$= 2 + 1$

$= 3$

0 1 1 2 3

a b c

$\cancel{0}$ $\cancel{1}$ $1\ 2$

$1\ 2$ 2

a = b;

a	b	c
$\cancel{2}$	$\cancel{3}$	$\cancel{5}$
3	5	8

```
int a=0, b=1;
int n=11;  int i=1;

while (i<=n) {
System.out.print(a+" ");

int c= a+b;

  a=b;
  b=c;
 i++;
}
```

$i = \cancel{1}\ \cancel{2}\ \cancel{3}\ \cancel{4}\ \cancel{5}\ \cancel{6} \ldots 11$

a	b	c
~~0~~	~~1~~	1
~~1~~	~~1~~	~~2~~
~~1~~	~~2~~	~~3~~
~~2~~	~~3~~	~~5~~
~~3~~	~~5~~	~~8~~
5	8	13

Output

0 1 1 2 3 5 8

13 21 34 55 89

144

Reverse of a number :-

```
int n=5678, reverseNo=0;

while(n>0) {

 int digit = n%10;
  n = n/10;
  reverseNo = reverseNo *10 + digit;
}
```

5678

8

8765

1221 ==

1221

```
if (n == reverseNo){
    Sout("Palindrome");
}
else {
    Sout("not Palindrome");
}
```

Print all divisors :-

(1234567899)

36 -> 1, 2, 3, 4, 6, 9, 12, 18, 36

int origN = n;

int i=1, n=36;

2^{32}

if(. == rev)

while(i<=n){

if(n%i==0)

2147

Sout(i+" ");

} i++;

GCD

Math.min(a,b);

(Common divisor)

(20, 40)

(16, 40) min(a,b)

(18, 40)

2,

2 ✓

4 ✓

8 ✓

```
int a=16, b=40;
int i=2, end = Math.min(a,b);
gcd=1;
while(i<=end) {
    if( a%i==0 && b%i==0) {
        gcd=i;
    }
    i++;
}
```

7, 17

(40, 80)

2

gcd = ~~2~~ ~~4~~ ~~16~~

~~20~~ 40